

Jayakumar Nagaraj

Mechanical Engineer

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PROFESSIONAL EXPERIENCE

Profile Summary:

Mechanical Engineering professional with 13+ years of experience across industrial gearbox design, wind turbine systems, and commercial vehicle rear axle systems. Proven track record in technical project management, design & development, and reliability engineering across organizations including Flender, Vestas, and Ashok Leyland. Skilled in cross-functional leadership, root cause analysis, and driving design improvements for complex drivetrain and structural systems, with a strong focus on performance, durability, and cost optimization.

Manager – Drive Train Engineering Product Development at Ashok Leyland Ltd, Chennai.

Jan 2026 – Present

Key Skill: *Cross-functional Leadership | Technical Problem Solving | Cost Optimization | RCA | Vehicle Testing & Validation | PPAP | APQP | NVH Basics*

Core Responsibilities:

- Act as a technical focal point for field and customer-reported issues related to rear axle systems across MHCV platforms.
- Analyze application-level failures and translate field concerns into engineering solutions ensuring performance and durability.
- Lead root cause analysis (RCA) and implement corrective actions in coordination with manufacturing, suppliers, and quality teams.
- Support cross-functional teams in resolving technical queries and ensuring alignment with customer requirements and usage conditions.
- Translated field failures into engineering action plans, ensuring design validation and production feasibility.
- Monitored validation activities (bench & vehicle testing) to ensure design changes meet performance and durability targets.
- Presented project status, risks, and mitigation plans to senior leadership in ER review (VP level).
- Delivered cost optimization and warranty reduction projects through structured execution and stakeholder alignment.

Technical Lead – Product Improvement at Vestas Wind System, (Consultant via ENABL Engineering Pvt Ltd)

Sep 2022 – Dec 2025

Key Skill: *Project Management | Product development | Problem Solving | PPAP | DMAIC | FMEA | Risk Assessment | Ansys | SAP | Windchill | Creo | SSGB*

Core Responsibilities:

- As a technical lead in Product Improvement team, I oversee multiple Wind Turbine CIM projects related to the Nacelle-Hub and powertrain module with the goal of minimizing the LPF.
- I have expertise in analyzing structural damage, identifying hydraulic anomalies, addressing tribological issues, and handling other mechanical systems.
- Specialized in investigating failure reports and Data analysis from field turbines; this involves close collaboration with WTG engineers and module engineers.
- Lead detailed root-cause analysis (RCA) of failure and identify technical and commercial impact on fleet.
- Coordinating with multiple stakeholders to conduct technical Root Cause Analysis (RCA) for wind turbine component issues. This includes addressing both the specific Technical and the Systemic RCA.
- Engaging with suppliers and manufacturers of affected components to address warranty claims, including the preparation of 8-D reports and cost recovery efforts.
- Executing Development and verification of the improvement solution. (Simulation, Validation and FFF tests at site).
- Drive the Engineering Change Management (ECM) process by authoring Change Requests/Notices (CR/CN) and ensure updates to DFMEA & PFMEA as a risk identification and mitigation process.
- Presenting technical Root Cause Analysis (RCA) and solutions in Gate meetings to the Program Director, enabling the progression through the respective stage Gates.
- Co-ordinate with the Supply chain team (SIOP) to ensure stock availability for projects as well as operational needs.
- Contribute to testing activities, including the development of test rigs, to validate solutions in collaboration with cross functional team.
- Gather key technical insights for customer and stakeholder communication while providing strategic inputs for business case development.

Key Skill: /Product Design/calculation/Kiss soft/ ST plus/ DFM/DFA / UG NX / Teamcenter/ SAP/ JIRA /SharePoint /GD&T /Leading projects/A3 problem solving/

Core Responsibilities:

- Led design order allocation, ensuring efficient workload distribution among team members.
- Handled design orders for helical and bevel gear drives across conveyors, cement mills, excavators, paper industries, and other applications.
- Conducted bending and fatigue safety calculations for gears per ISO 6336/DIN 3990 standards.
- Collaborated with multiple stakeholders to identify root causes and provide solutions for customer/field issues.
- Designed and developed gearbox components, including housing, gears, shafts, and bearings.
- Reviewed and approved general arrangement drawings for standard and non-standard gearboxes per customer requirements.
- Expert in UG NX, with hands-on experience in top-down assembly, skeleton modeling, layout design, piping, dimensional drawings, corrosion prevention, and spare parts documentation.
- Prepared BOM for gearboxes and auxiliaries using SAP PF1 and Teamcenter.
- Designed casting structures, housings, and machining drawings for industrial and wind applications.
- Created material master for new parts in SAP PF1 and tracked processes via JIRA.
- Applied strong GD&T knowledge (ASME Y14.5M) for assembly, including stack-up analysis.
- Executed bearing selection calculations for gearboxes and various applications.

Product Development Engineer at Woosu Automotive India Pvt Ltd.,

June 2012 - Mar 2014

Core Responsibilities:

- Managed the development schedule and coordinated follow-ups with new suppliers to ensure timely project execution.
- Prepared Bill of Materials (BOM) for new product development, optimizing material selection and cost efficiency.
- Conducted Failure Modes and Effects Analysis (FMEA) and developed Control Plans, Process Flow Diagrams (PFD), and Standard Operating Procedures (SOP).
- Oversaw documentation processes related to Engineering Change Orders (ECO) and drawing management, ensuring accuracy and compliance.
- Led ECO reviews with customers, facilitating discussions and implementing necessary drawing improvements.
- Executed ECO changes using AutoCAD and SAP, ensuring seamless integration of design modifications into production workflows.

Accomplishments:

- Awarded “Best engineer – Quality (98.3% in 2020)” on the projects handled with Flender gearboxes.
- Led the localization project of Industrial gear parts from German base to India under local Engineering and implemented.
- Delivered fixture design for planet carrier, torque arm and Housing components with 30% cost saving.
- Successfully addressed cavitation issues in the HPU internal gear pump and implemented an effective solution using a modified suction pipe across the entire 6MW fleet.
- Successfully resolved Pitch cylinder seal leaking issues, identified potential root causes, and implemented improved solutions across the 4MW fleet in the LATAM region.
- Retained Key customer by delivering technical solutions to customer complaints and expectations.
- **International Collaboration:**
 - Travelled onsite visit to a supplier facility in Spain to validate the test rig for the pitch cylinder project. Addressed and resolved vibration issues in the test rig with supplier involvement during the visit.

Academic Synopsis:

- **M. Tech in Design Engineering** – First class with 7.4 CGPA from BITS Pilani WILP, Rajasthan in May 2025.
- **B.E. Mechanical Engineering** - First class with 7.8 CGPA from SACS MAVMM Engineering, Madurai in May 2012.

Certifications:

- **PTC Creo** – Advanced design - Certified in Creo elements.
- **Six Sigma Green belt** - Certified in process optimization, statistical analysis and quality improvement methodologies.

Activities

- Member of recruitment panel
- 5S Team member
- ERT Member

Traits

- Lead capable
- Quality motive
- Easy adaptive

Profile

- Home City - Udumalaipettai
- Living City - Chennai
- Languages - Tamil, English, Kannada, German (A1)